More Circuits: Ammeters & Voltmeters

Goal.

- Meet two measuring devices — the ammeter and the voltmeter — and figure out from first principles what resistance each one must have.
- See that the loop rule is bigger than "find the current": the same equation lets you solve for *whatever single thing* you don't already know, once a meter tells you something new.
- Work three versions of one loop, each with a different unknown.

1. Ideal meters

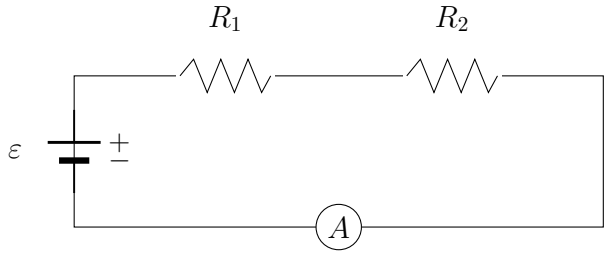
Ammeter: measures _____ Inserted _____ in the circuit (break the wire, wire the meter into the gap) so the same current that was already flowing still flows straight through it. An ideal ammeter has resistance _____

Voltmeter: measures the _____ between two points. Connected _____ across an element (its own branch, tapping both ends — it is never spliced into the main wire). An ideal voltmeter has resistance _____

Check. An ammeter must not change the current it's reporting; a voltmeter must not divert any current away from the branch it's tapping. Given that, which meter needs zero resistance, and which needs (essentially) infinite resistance?

2. Problem 1 — the current

A 9.0 V ideal battery drives current through $R_1 = 1.8\ \text{k}\Omega$ and $R_2 = 2.7\ \text{k}\Omega$ in series. An ammeter is wired into the loop as shown.



Write the loop rule for this circuit and solve for the current the ammeter reads.

start from the loop rule: $\varepsilon - IR_1 - IR_2 = 0$

$I =$

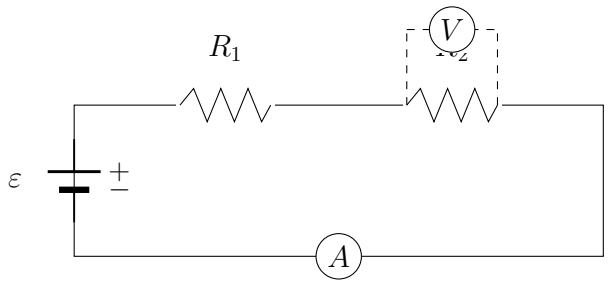
Now with the numbers above:

$I =$

Check. The ammeter sits on the bottom wire, not between R_1 and R_2. Would it read something different if you moved it anywhere else in this loop? Why?

3. Problem 2 — a voltmeter's reading

Same loop, but now only $R_1 = 2.0\ \text{k}\Omega$ and the EMF ($\varepsilon = 12\ \text{V}$) are given — R_2's resistance is unknown. The ammeter reads $I = 3.0\ \text{mA}$, and a voltmeter is connected across R_2.



Use the loop rule to predict the voltmeter's reading. (You will not need R_2's resistance for this part.)

start from the loop rule: $\varepsilon - IR_1 - \Delta V_2 = 0$

$\Delta V_2 =$

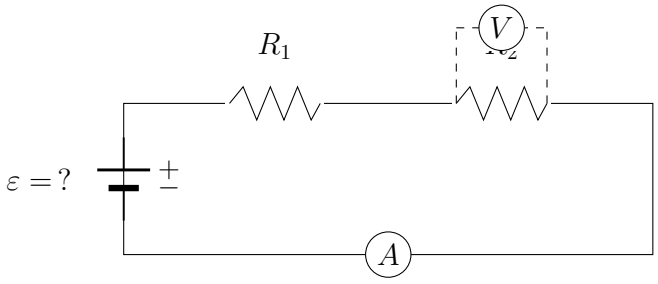
Now with the numbers above:

$\Delta V_2 =$

Check. You found the voltmeter's reading without ever using R_2's resistance. Why didn't you need it?

4. Problem 3 — the EMF itself

Same loop again, but now the battery's EMF is what's unknown. $R_1 = 2.0\ \text{k}\Omega$ is given, the ammeter reads $I = 4.0\ \text{mA}$, and the voltmeter across R_2 reads 5.0 V.



Use the loop rule to find ε.

start from the loop rule: $\varepsilon - IR_1 - \Delta V_2 = 0$

$\varepsilon =$

Now with the numbers above:

$\varepsilon =$

Check. Using I and the voltmeter reading, could you now find R_2 too, even though its resistance was never given directly? What is it?

5. Two loops, two batteries

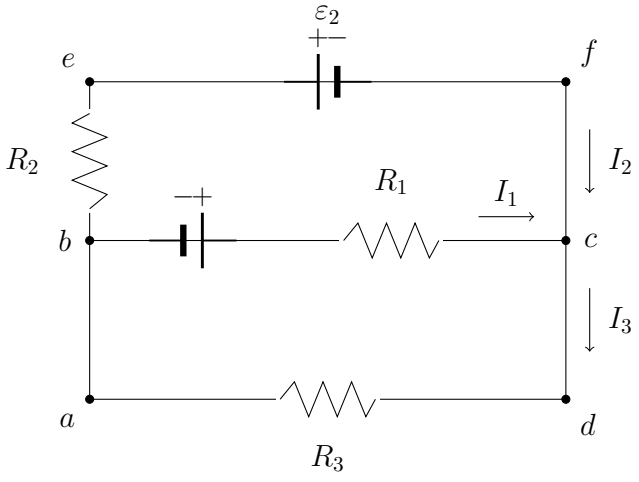
A single loop only ever gives you *one* equation. The circuit below has three branches meeting at b and at c, so there are three unknown currents I_1, I_2, I_3 — one loop equation can't pin down three unknowns.

How many equations do you need, and where does each one come from? _____

Check. Why doesn't the loop rule applied to the *outer* perimeter (e-f-d-a-e) give you a fourth, independent equation?

6. Problem 4 — all three currents

In the circuit below, $\varepsilon_1 = 9.0$ V, $R_1 = 4\ \Omega$, $\varepsilon_2 = 16$ V, $R_2 = 5\ \Omega$, $R_3 = 2\ \Omega$.



Write the loop rule for each loop and the junction rule at c, then solve the resulting system for all three currents.

Loop 1 (through ε_2, R_2): $I_1R_1 - I_2R_2 = \varepsilon_1 + \varepsilon_2$ Loop 2 (through R_3): $I_1R_1 + I_3R_3 = \varepsilon_1$ Junction at c: $I_1 + I_2 = I_3$

$I_1 =$

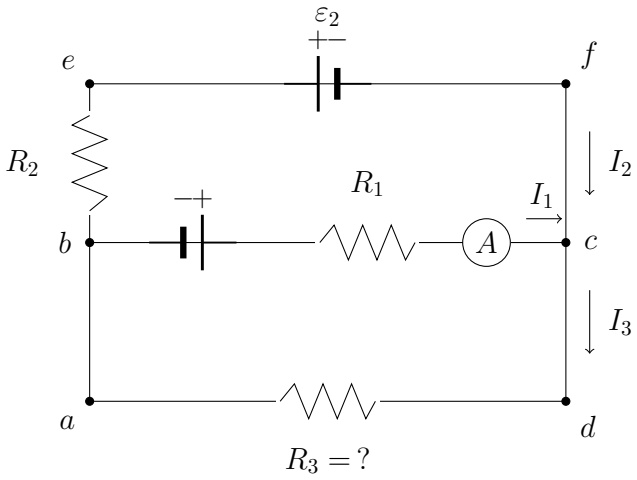
$I_2 =$

$I_3 =$

Check. One of your currents came out negative. What does that tell you about the direction you assumed for it on the diagram?

7. Problem 5 — an unknown resistor

Same shape, new numbers: $\varepsilon_1 = 20$ V, $R_1 = 3\ \Omega$, $\varepsilon_2 = 10$ V, $R_2 = 5\ \Omega$. This time R_3 is unknown — but an ammeter in the ε_1-R_1 branch reads $I_1 = 4.0$ A.



Use the loop rule and the junction rule to find R_3. (Work out I_2, then I_3, on your way there.)

Loop 1 gives I_2; the junction rule gives I_3; Loop 2 ($\varepsilon_1 = I_1 R_1 + I_3 R_3$) then gives R_3

$I_2 =$

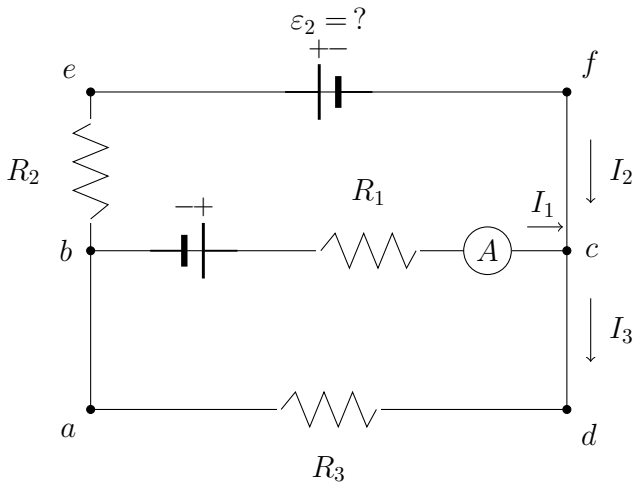
$I_3 =$

$R_3 =$

Check. Could you have found R_3 without first finding I_2? Why or why not?

8. Problem 6 — an unknown EMF

Same shape again: $R_1 = 5\ \Omega$, $R_2 = 6\ \Omega$, $R_3 = 4\ \Omega$. This time ε_2 is unknown, but $\varepsilon_1 = 21$ V is given, and the ammeter reads $I_1 = 3.0$ A.



Find ε_2. (This time work out I_3 first, then I_2.)

Loop 2 ($\varepsilon_1 = I_1 R_1 + I_3 R_3$) gives I_3; the junction rule gives I_2; Loop 1 then gives ε_2

$I_3 =$

$I_2 =$

$\varepsilon_2 =$

Check. Compare the order you worked in for Problems 5 and 6 — which loop did you need *first* in each, and why did the unknown force that order?